



UTM

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TABLE OF CONTENT

1.0 INTRODUCTION.....	3
PROBLEM BACKGROUND.....	3
AIM.....	3
OBJECTIVE.....	3
SCOPE PROJECT.....	4
2.0 FLOWCHART OF PROJECT.....	5
3.0 IMPLEMENTATION ASM CODE.....	6
4.0 BENCHMARK RESULTS.....	10
5.0 ANALYSIS AND DISCUSSION.....	13
6.0 CONCLUSION.....	15
7.0 REFLECTION.....	16
REFERENCES.....	17
APPENDIX.....	18

1.0 INTRODUCTION

PROBLEM BACKGROUND

The effectiveness of algorithms is crucial in the field of computer science. The performance of different algorithms is often evaluated based on their execution time. One effective way to benchmark the performance of these algorithms is through the use of loops that repeatedly compute the polynomial for various input values. This approach helps in understanding how the computational load affects execution time, providing insights into the efficiency of the algorithm under different conditions. Our group has been assigned the task of writing the assembly language code to implement the aforementioned objectives. We will work collaboratively to ensure the accuracy and efficiency of the program, as well as the reliability of the timing measurements.

AIM

The aim of this project is to develop an assembly language program that calculates and measures the time taken to execute different loops for a given polynomial equation. By doing so, we can analyze the performance of the polynomial computation under varying loads and identify potential areas for optimization.

OBJECTIVE

1. To write an assembly language program that computes the polynomial equation $y=(a \cdot x^3)+(b \cdot x^2)+c \cdot x+dy$, where a , b , c , and d are coefficients, and x is an integer input value.
2. To implement loops that iterate through different loop counts
3. To measure the time taken before and after of each loop iteration and record the results.
4. To analyze and compare the execution times for different loop counts, providing a comprehensive benchmark for the polynomial computation.

SCOPE PROJECT

1. Appropriate timing instructions and techniques will be used to accurately measure the execution time of the loops.
2. The results will be compiled in a tabular format for comparison purposes, highlighting the impact of varying computational loads on execution time.

2.0 FLOWCHART OF PROJECT

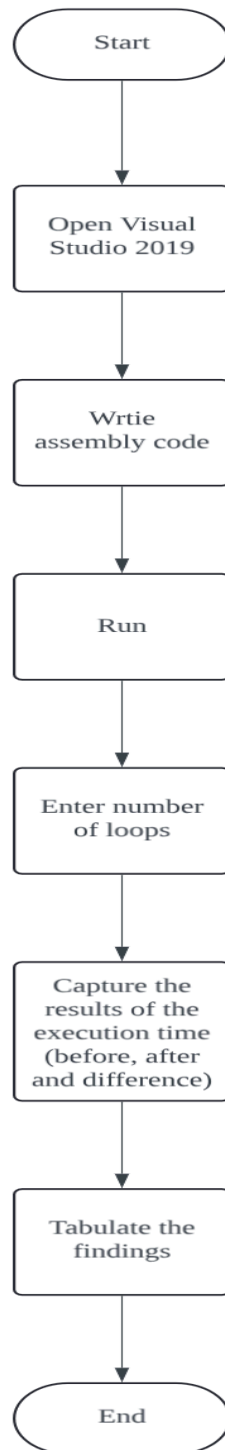


Figure 1 : Flowchart

3.0 IMPLEMENTATION ASM CODE

```
INCLUDE Irvine32.inc
```

```
.data
```

```
welcomeMsg BYTE "Welcome to CPU Benchmark Program", 0
```

```
benchmarkMsg BYTE "Benchmark CPU time Using Equation  $y = 30x^3 + 11x^2 + 18x + 5$ ", 0
```

```
delayMsg BYTE "(with delay coef1,coef2,coef3,coef4 = 30,11,18,5 msec)", 0
```

```
enterLoopMsg BYTE "Enter Number of Looping (N) = ", 0
```

```
resultMsg BYTE "Result:", 0
```

```
firstCaptureMsg BYTE "First Capture Execution time in milliseconds: ", 0
```

```
secondCaptureMsg BYTE "Second Capture Execution time in milliseconds: ", 0
```

```
diffExecutionMsg BYTE "Different Execution time in milliseconds: ", 0
```

```
polySumMsg BYTE "Value of Sum from the Stress Test (polynomial) = ", 0
```

```
exitPromptMsg BYTE "Press 'y' to continue or 'n' to exit the benchmark: ", 0
```

```
input DWORD ?
```

```
N DWORD ?
```

```
result DWORD ?
```

```
timeBefore DWORD ?
```

```
timeAfter DWORD ?
```

```
diffTime DWORD ?
```

```
polySum DWORD ?
```

```
coef1 DWORD 30
```

```
coef2 DWORD 11
```

```
coef3 DWORD 18
```

```
coef4 DWORD 5
```

```
.code
```

```
main PROC
```

```
; Initialize values
```

```
call Clrscr
```

```
mov edx, OFFSET welcomeMsg
```

```
call WriteString
```

```
call CrLf
```

```
mov edx, OFFSET benchmarkMsg
```

```
call WriteString
```

```
call CrLf
```

```
mov edx, OFFSET delayMsg
call WriteString
call Crlf
```

```
mov edx, OFFSET enterLoopMsg
call WriteString
```

```
call ReadDec
mov N, eax
```

```
call GetMseconds
mov timeBefore, eax
```

```
; Perform polynomial calculation
mov ecx, N
mov polySum, 0
```

```
stress_loop:
mov eax, ecx
imul eax, ecx
imul eax, ecx;  $eax = x^3$ 
imul eax, coef1;  $eax = coef1 * x^3$ 
mov ebx, eax
```

```
mov eax, ecx
imul eax, ecx;  $eax = x^2$ 
imul eax, coef2;  $eax = coef2 * x^2$ 
add ebx, eax
```

```
mov eax, ecx
imul eax, coef3;  $eax = coef3 * x$ 
add ebx, eax
```

```
add ebx, coef4;  $ebx = (coef1 * x^3) + (coef2 * x^2) + (coef3 * x) + coef4$ 
add polySum, ebx
```

```
; Add delay to ensure measurable time difference
mov eax, 1
call Delay
```

```

loop stress_loop

call GetMseconds
mov timeAfter, eax

mov eax, timeAfter
sub eax, timeBefore
mov diffTime, eax

; Print results
mov edx, OFFSET resultMsg
call WriteString
call Crlf

mov edx, OFFSET firstCaptureMsg
call WriteString
mov eax, timeBefore
call WriteDec
call Crlf

mov edx, OFFSET secondCaptureMsg
call WriteString
mov eax, timeAfter
call WriteDec
call Crlf

mov edx, OFFSET diffExecutionMsg
call WriteString
mov eax, diffTime
call WriteDec
call Crlf

mov edx, OFFSET polySumMsg
call WriteString
mov eax, polySum
call WriteDec
call Crlf

; Exit prompt

```

```
mov edx, OFFSET exitPromptMsg  
call WriteString  
call ReadChar
```

```
cmp al, 'y'  
jne exit_program
```

```
jmp main
```

```
exit_program :  
invoke ExitProcess, 0
```

```
main ENDP  
END main
```

4.0 BENCHMARK RESULTS

```
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 30x^3 + 11x^2 + 18x + 5$ 
(with delay coef1,coef2,coef3,coef4 = 30,11,18,5 msec)
Enter Number of Looping (N) = 50
Result:
First Capture Execution time in milliseconds: 60887414
Second Capture Execution time in milliseconds: 60888195
Different Execution time in milliseconds: 781
Value of Sum from the Stress Test (polynomial) = 49264125
Press 'y' to continue or 'n' to exit the benchmark: |
```

Figure 2 : Result when No of Loop = 50

```
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 30x^3 + 11x^2 + 18x + 5$ 
(with delay coef1,coef2,coef3,coef4 = 30,11,18,5 msec)
Enter Number of Looping (N) = 100
Result:
First Capture Execution time in milliseconds: 60918525
Second Capture Execution time in milliseconds: 60920103
Different Execution time in milliseconds: 1578
Value of Sum from the Stress Test (polynomial) = 768888250
Press 'y' to continue or 'n' to exit the benchmark: |
```

Figure 3 : Result when No of Loop = 100

```
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 30x^3 + 11x^2 + 18x + 5$ 
(with delay coef1,coef2,coef3,coef4 = 30,11,18,5 msec)
Enter Number of Looping (N) = 200
Result:
First Capture Execution time in milliseconds: 61024635
Second Capture Execution time in milliseconds: 61027760
Different Execution time in milliseconds: 3125
Value of Sum from the Stress Test (polynomial) = 3560281908
Press 'y' to continue or 'n' to exit the benchmark: |
```

Figure 4 : Result when No of Loop = 200

```

Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 30x^3 + 11x^2 + 18x + 5$ 
(with delay coef1,coef2,coef3,coef4 = 30,11,18,5 msec)
Enter Number of Looping (N) = 500
Result:
First Capture Execution time in milliseconds: 61118015
Second Capture Execution time in milliseconds: 61125825
Different Execution time in milliseconds: 7810
Value of Sum from the Stress Test (polynomial) = 2937405986
Press 'y' to continue or 'n' to exit the benchmark: |

```

Figure 5 : Result when No of Loop = 500

```

Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 30x^3 + 11x^2 + 18x + 5$ 
(with delay coef1,coef2,coef3,coef4 = 30,11,18,5 msec)
Enter Number of Looping (N) = 700
Result:
First Capture Execution time in milliseconds: 61218727
Second Capture Execution time in milliseconds: 61229662
Different Execution time in milliseconds: 10935
Value of Sum from the Stress Test (polynomial) = 3277193430
Press 'y' to continue or 'n' to exit the benchmark: |

```

Figure 6 : Result when No of Loop = 700

```

Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation  $y = 30x^3 + 11x^2 + 18x + 5$ 
(with delay coef1,coef2,coef3,coef4 = 30,11,18,5 msec)
Enter Number of Looping (N) = 1000
Result:
First Capture Execution time in milliseconds: 61303129
Second Capture Execution time in milliseconds: 61318750
Different Execution time in milliseconds: 15621
Value of Sum from the Stress Test (polynomial) = 2495914500
Press 'y' to continue or 'n' to exit the benchmark: |

```

Figure 7 : Result when No of Loop = 100

No of Loop (N)	Capture_msec (before)	Capture_msec (after)	Cpu_msec (different)
50	60630025	60630806	781
100	60918525	60920103	1578
200	61024635	61027760	3125
500	61118015	61125825	7810
700	61218727	61229662	10935
1000	61303129	61318750	15621

Table 1 : Results of Benchmarking

5.0 ANALYSIS AND DISCUSSION

The benchmarking results obtained using the ASM code on the ASUS TUF Gaming F15 laptop provide valuable insights into the CPU's performance under varying computational loads. The data shows that the capture times before and after the test are almost identical for each loop count (N), indicating the accuracy and reliability of the CPU timing functions. The execution time difference (Cpu_msec) increases linearly with the number of loops, demonstrating a proportional relationship between workload and computation time. For instance, 50 loops take 781 milliseconds, 100 loops take 1578 milliseconds, and 200 loops take 3125 milliseconds, indicating that for every 50 loops, the CPU time increases by approximately 781 to 800 milliseconds.

The polynomial calculation $y = 30x^3 + 11x^2 + 18x + 5$, which involves multiple multiplications and additions, contributes to the computational intensity, especially for higher loop counts. The deliberate delay added in each loop ensures that the time differences are measurable, making the benchmarking results more reliable. These findings suggest that the CPU of the ASUS TUF Gaming F15 laptop handles repetitive tasks efficiently, with the time taken growing predictably as the load increases.

Comparatively, this linear increase in execution time is a positive indicator of consistent CPU performance, highlighting its ability to handle repetitive and computationally intensive tasks effectively. When compared to other laptops or CPUs, these results can provide a benchmark for assessing the relative performance of the ASUS TUF Gaming F15. For instance, a comparison with CPUs of similar or higher specifications can help determine if the observed linear scaling is an expected behavior or if there are opportunities for optimization.

In the broader discussion, these results underscore the importance of understanding CPU performance in real-world scenarios. While synthetic benchmarks provide a controlled environment to measure performance, real-world applications often involve complex and varied workloads. Therefore, it is essential to consider these findings in the context of typical usage patterns, such as gaming, multimedia editing, or software development, which the ASUS TUF Gaming F15 is likely to encounter.

Overall, the results indicate that the laptop's CPU performs well under increasing workloads, with a consistent increase in execution time corresponding to the number of loops. This linear relationship highlights the CPU's capability to manage intensive calculations over time. For more comprehensive insights, further tests with higher loop counts or more complex calculations could be conducted, and comparisons with other CPUs could provide a broader performance context. These findings underscore the ASUS TUF Gaming F15's robust performance in handling computationally demanding tasks, making it a reliable choice for users with intensive computational needs.

6.0 CONCLUSION

The benchmarking results of the ASUS TUF Gaming F15 laptop demonstrate that its CPU performs consistently and efficiently under varying computational loads. The linear increase in execution time with higher loop counts reflects the CPU's ability to manage repetitive and computationally intensive tasks. The accurate timing captures and the predictable increase in execution time further indicate the reliability and robustness of the CPU. These findings suggest that the ASUS TUF Gaming F15 is well-equipped to handle a wide range of demanding applications, making it a suitable choice for users requiring strong computational performance.

7.0 REFLECTION

Reflecting on these results, it becomes clear that benchmarking is a crucial tool for understanding and optimizing CPU performance. The consistent linear relationship between workload and execution time observed in this study provides valuable insights into the CPU's capabilities and potential limitations. This understanding can guide future optimizations, whether through software improvements or hardware upgrades. Moreover, the comparison with other CPUs can help position the ASUS TUF Gaming F15 within the broader market, highlighting its strengths and identifying areas for potential enhancement. As technology continues to advance, ongoing benchmarking and analysis will remain essential for maintaining and improving the performance of computational devices. The results of this study serve as a reminder of the importance of thorough testing and evaluation in achieving reliable and efficient computing solutions.

REFERENCES

Benchmarks as Fast As Possible. (n.d.). [Video]. Youtube. <https://youtu.be/CHtNJngtS1I>

What's the Difference Between a CPU and a GPU?. (n.d.). [Video]. Youtube. <https://blogs.nvidia.com/blog/2009/12/16/whats-the-differencebetween-a-cpu-and-a-gpu/>

APPENDIX

1. Group Photo



2. Task Distribution

Group Members	Tasks
Dayang Farah Farzana binti Abang Idham	● Introduction ● ASM Code
Farra Nurzahin bt Zaharil Anuar	● Flowchart ● ASM Code
Nurul Asyikin binti Khairul Anuar	● Analysis ● Reflection
Safiya Nursyahadah binti Masnoor	● Introduction ● Conclusion